Osanda Wedamulla

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Experienced Lead Software Engineer/Architect with over **10 years** in the industry, specializing in **scalable** application development and **system architecture**. Proven leader in **optimizing** software processes, **automating** operations, and delivering **high-performance** enterprise solutions. Passionate about technology **innovation**, with expertise in various programming languages and frameworks to drive **impactful** results.

Key Achievements

- Achieved a 90% increase in productivity by automating repetitive tasks, including testing, deployment, and monitoring, while streamlining CI/CD pipelines.
- Implemented efficient logging and monitoring solutions, providing real-time insights into application behavior and state with 0% decrease with time in terms customer support response and troubleshooting.
- Introduced industry best practices in security, scalability, code optimization, and DevOps resulting in 40% improved code quality, system reliability, and overall performance of deployed applications.
- Collaborated with DevOps teams to establish automated deployment workflows, achieving zero-downtime deployments and improving release frequency.
- Reduced deployment time by 80% through Docker containerization and Kubernetes automation.
- Led the implementation of comprehensive monitoring and alerting using Prometheus and Grafana, achieving 90% faster incident response time and improved system reliability.
- Architected and optimized CI/CD pipelines with GitLab and Github Actions to implement GitOps practices with Argo CD, reducing deployment time by 30% and enabling seamless rollbacks effective integration.
- Designed and deployed cloud-native applications deployment and maintenance on GCP/AWS utilizing GKE/EKS, optimizing resource utilization and cost efficiency by 40%.
- Developed customized Ansible modules and roles with Python for streamlined automation of repetitive workflows.

Technical Skills

- **Programming Languages:** Java (8+ years), JavaScript/TypeScript (4 years), Golang (3 years), Python (4+ years), Kotlin (2 years), Shell/Bash Scripting (5+ years)
- **Frameworks & Development:** Spring MVC/Boot/Cloud (7+ years), Django/FastAPI (2 years), Apache Kafka/RabbitMQ (5 years), REST/GraphQL (8+ years), Reactive/Functional Programming (2 years)
- **Databases & Storage:** MySQL/MariaDB/PostgreSQL (7+ years), MongoDB/Redis (5+ years), Apache Cassandra/AstraDB (2 years), RDS/DynamoDB (4 years)
- **Cloud & DevOps:** AWS (EC2, ECS, VPC, IAM, S3, Lambdas) (5 years), GCP/Azure/DigitalOcean/Linode (3 years), Docker/Kubernetes/Helm (5+ years), Kubernetes EKS/GKE/AKS (5+ years), Ansible/Terraform (3 years), ArgoCD Automations (2 years), Minio/Longhorn Storage (4 years)
- **Monitoring & Security:** NewRelic/DataDog (4 years), ELK/EFK Stack (4 years), Prometheus/Grafana (5+ years), GitHub Actions (5+ years), Hasicorp Vault (5+ years), SSL/TLS Management (4 years), IAM/RBAC (5+ years)
- **Version Control & Methodologies:** Git (GitHub, GitLab, Bitbucket) (5+ years), Agile (Scrum/Kanban) (5+ years), Database Migration (Flyway/Liquibase) (4 years), AI Agents/Prompt Engineering (4 years)

Experience

V Fullstack GmbH | Senior Software Engineer - Team Lead | April 2021 - Present (Remote)

- Provided technical leadership and mentorship to engineering teams, guiding architectural decisions and fostering a culture of engineering excellence and continuous improvement.
- Conducted code reviews and design critiques, establishing quality standards that improved code maintainability

and reduced technical debt by 40%.
- Integrated SonarQube, Checkstyle, and PMD-based static code analyzers in CI/CD pipelines to enforce best code practices and maintain code quality.
- Architected and led the development of a high-throughput microservices ecosystem using Spring Boot and Kafka, handling 10K+ transactions per minute with 99.9% uptime and 100ms average response time.
- Orchestrated database optimization initiatives across PostgreSQL and MongoDB, reducing query execution time by 75% and implementing efficient indexing strategies.
- Architected and implemented scalable microservices architecture supporting a large number of services, ensuring high availability (99.9% uptime) and optimal performance for enterprise-level applications.
- Developed a multi-tenant authentication system using Spring Security and Keycloak, supporting OAuth2 and JWT for secure API access across 20+ microservices.
- Deployed and managed applications on managed/on-premises Kubernetes clusters utilizing Helm Charts and ArgoCD for efficient automated deployment and maintenance of production-ready applications.
- Led the implementation of comprehensive monitoring and alerting using Prometheus and Grafana, achieving 90% faster incident response time and improved system reliability.
- Built and maintained Python FastAPI services for ML model deployment, achieving 90% reduction in model inference time and seamless integration with existing microservices.
- Maintained, managed, and monitored Kubernetes clusters and related applications and services using tools like Prometheus, Grafana, ISTIO and Linkerd.

Remote Sensing Metrics Asia (Pvt) Ltd. | Software Engineer / Senior Software Engineer | December 2015 - March 2021

- Led technical design reviews and provided architectural oversight for critical features, ensuring adherence to non-functional requirements including scalability, security, and maintainability.
- Developed and maintained dynamic web applications by integrating multiple data feeds from various sources using technologies such as Spring Boot, Java, Angular, MySQL, PostgreSQL, MongoDB, and Redis.
- Utilized Spring Security, OAuth2, Keycloak, and other security adapters to effectively manage access control across applications. Implemented a centralized authentication server using JBoss/Keycloak with Spring Boot adapters for secure hosting on AWS servers.
- Demonstrated deep hands-on experience in requirement analysis, software design and architecture, implementation, testing, management, and monitoring of complex software systems with stringent requirements.
- Oversaw the migration process of all applications to the cloud environment and successfully resolved production issues that arose during the transition.
- Wrote unit and integration test cases with JUnit, Mockito, MockMVC, Test Containers, Spring MVC Test, Spring Boot Test, and Karate testing framework.
- Conducted code reviews and design critiques, establishing quality standards that improved code maintainability and reduced technical debt by 60%.
- Identifying technology limitations and deficiencies in existing systems and associated processes, procedures and methods to overcome project development related issues.
- Integrated Apache Kafka and RabbitMQ messaging with Spring Boot applications for efficient real-time data processing following event-driven architecture.
- Implemented container orchestration strategies using Kubernetes, managing multi-environment deployments across development, staging, and production clusters.

Education

Professional Qualifications and Certifications

- KodeKloud Devops Certificates (April 2024 - May 2025)
    - Kubernetes and Cloud Native Security Associate (KCSA) - Certificate
    - GitOps with ArgoCD - Certificate

Academic Education

- Bachelor of Science in Computer Science at Sabaragamuwa University of Sri Lanka (2011 - 2015)